

## KNOX COUNTY AP, OH, USA

WMO: 720414

Lat: 40.333N

Lon: 82.517W

Elev: 1191

StdP: 14.07

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00139

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 1.0        | 7.1 | -4.4                          | 4.5 | 4.5  | 0.0 | 5.7 | 8.5  | 25.1                  | 28.1 | 23.0 | 26.9 | 5.7                   | 270  | 0.464 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 89.8            | 74.4 | 87.6 | 73.3 | 84.4 | 71.7 | 77.3                | 85.2 | 75.4 | 83.5 | 74.1 | 81.2 | 6.9                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.8                            | 136.3 | 81.4 | 73.0 | 127.9 | 79.1 | 72.2 | 124.6 | 78.3 | 41.5          | 84.5 | 39.9 | 83.2 | 38.5 | 80.8 | 82.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| 20.6              | 18.2 | 16.1 | -5.1                       | 92.8 | 6.6                | 3.2 | -9.9                                               | 95.0 | -13.7      | 96.9 | -17.4      | 98.6 | -22.2      | 100.9 |    |    |
|                   |      |      | -5.5                       | 79.4 | 6.0                | 1.8 | -9.8                                               | 80.7 | -13.3      | 81.7 | -16.7      | 82.7 | -21.1      | 84.0  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.9   | 26.4  | 30.6  | 39.5  | 52.0  | 63.2  | 70.0  | 73.3  | 71.7  | 66.1  | 53.8  | 42.2  | 33.2  |
|                                                                        | DBStd       | 18.63  | 11.67 | 11.92 | 10.81 | 9.27  | 8.19  | 5.81  | 4.98  | 4.62  | 7.30  | 9.07  | 9.72  | 10.09 |
|                                                                        | HDD50       | 2597   | 735   | 553   | 359   | 85    | 5     | 0     | 0     | 0     | 1     | 61    | 271   | 527   |
|                                                                        | HDD65       | 5626   | 1197  | 964   | 791   | 400   | 135   | 23    | 3     | 6     | 74    | 362   | 686   | 985   |
|                                                                        | CDD50       | 3305   | 2     | 9     | 33    | 144   | 414   | 599   | 724   | 672   | 485   | 180   | 36    | 7     |
|                                                                        | CDD65       | 860    | 0     | 0     | 1     | 9     | 79    | 172   | 261   | 213   | 108   | 16    | 1     | 0     |
|                                                                        | CDH74       | 6874   | 0     | 0     | 6     | 94    | 664   | 1324  | 2161  | 1613  | 879   | 130   | 3     | 0     |
|                                                                        | CDH80       | 2055   | 0     | 0     | 1     | 9     | 143   | 385   | 760   | 451   | 280   | 26    | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 6.7    | 8.2   | 8.1   | 8.2   | 8.4   | 6.7   | 5.9   | 4.6   | 4.5   | 5.1   | 6.6   | 7.3   | 7.6   |
| (15) Precipitation                                                     | PrecAvg     | 41.3   | 3.1   | 2.2   | 3.2   | 3.9   | 4.4   | 4.2   | 4.3   | 3.9   | 3.4   | 2.7   | 2.8   | 3.3   |
|                                                                        | PrecMax     | 55.7   | 8.9   | 5.0   | 6.2   | 6.5   | 8.3   | 9.3   | 10.2  | 7.3   | 7.5   | 5.6   | 5.4   | 8.1   |
|                                                                        | PrecMin     | 28.1   | 0.8   | 1.1   | 1.3   | 1.9   | 1.8   | 0.7   | 1.7   | 1.4   | 1.3   | 0.7   | 0.9   | 1.4   |
|                                                                        | PrecStd     | 6.2    | 1.8   | 1.0   | 1.3   | 1.1   | 2.0   | 1.9   | 2.2   | 1.7   | 1.4   | 1.4   | 1.0   | 1.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 60.5  | 66.4  | 73.3  | 81.3  | 86.5  | 90.5  | 93.1  | 90.3  | 90.5  | 83.7  | 72.0  | 63.0  |
|                                                                        |             | MCWB   | 57.0  | 57.8  | 60.6  | 61.7  | 70.7  | 74.0  | 77.8  | 73.0  | 74.9  | 71.4  | 59.1  | 58.8  |
|                                                                        | 2%          | DB     | 54.8  | 58.8  | 66.4  | 76.9  | 83.8  | 87.8  | 90.0  | 87.7  | 87.6  | 77.2  | 65.8  | 57.2  |
|                                                                        |             | MCWB   | 51.5  | 53.9  | 56.7  | 60.2  | 68.3  | 72.4  | 75.7  | 73.0  | 72.6  | 64.2  | 55.9  | 54.0  |
|                                                                        | 5%          | DB     | 48.3  | 54.1  | 61.4  | 72.3  | 81.2  | 84.2  | 87.6  | 84.4  | 82.4  | 73.0  | 61.4  | 53.8  |
|                                                                        |             | MCWB   | 46.1  | 50.4  | 53.2  | 58.6  | 67.3  | 71.0  | 74.4  | 71.6  | 70.9  | 62.2  | 52.7  | 51.8  |
|                                                                        | 10%         | DB     | 43.2  | 47.9  | 56.9  | 67.9  | 78.5  | 81.6  | 84.2  | 82.0  | 79.2  | 68.5  | 57.0  | 48.1  |
|                                                                        |             | MCWB   | 39.8  | 44.1  | 50.2  | 56.3  | 65.5  | 69.2  | 72.7  | 70.4  | 68.4  | 61.6  | 50.1  | 46.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 57.2  | 60.1  | 62.7  | 66.1  | 74.6  | 77.1  | 80.2  | 78.1  | 76.6  | 72.4  | 62.8  | 59.5  |
|                                                                        |             | MCDB   | 59.8  | 64.8  | 72.6  | 74.9  | 82.6  | 86.6  | 88.4  | 85.5  | 86.2  | 78.6  | 68.4  | 61.8  |
|                                                                        | 2%          | WB     | 52.4  | 55.0  | 58.9  | 62.9  | 71.7  | 74.7  | 78.1  | 76.0  | 74.8  | 68.6  | 58.6  | 54.9  |
|                                                                        |             | MCDB   | 54.3  | 57.0  | 65.2  | 70.7  | 79.8  | 83.0  | 85.9  | 83.0  | 83.7  | 73.5  | 63.1  | 57.0  |
|                                                                        | 5%          | WB     | 46.7  | 50.2  | 55.4  | 60.5  | 69.4  | 73.2  | 76.3  | 74.3  | 73.0  | 65.5  | 54.3  | 51.5  |
|                                                                        |             | MCDB   | 47.5  | 54.2  | 59.7  | 68.6  | 76.8  | 81.4  | 84.3  | 80.5  | 80.4  | 70.5  | 60.2  | 53.3  |
|                                                                        | 10%         | WB     | 39.9  | 44.3  | 51.7  | 58.4  | 67.5  | 71.7  | 74.4  | 72.8  | 71.1  | 62.1  | 51.1  | 45.6  |
|                                                                        |             | MCDB   | 42.8  | 47.6  | 56.0  | 66.0  | 74.9  | 79.3  | 81.6  | 78.8  | 77.0  | 67.8  | 56.1  | 47.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.5  | 15.5  | 17.4  | 20.3  | 20.4  | 19.2  | 19.3  | 20.0  | 19.9  | 18.9  | 16.1  | 12.7  |
|                                                                        |             | MCDBR  | 19.4  | 21.6  | 23.8  | 25.3  | 23.0  | 22.4  | 21.7  | 22.1  | 23.6  | 24.4  | 22.1  | 18.5  |
|                                                                        |             | MCWBR  | 17.6  | 17.2  | 15.6  | 14.4  | 11.2  | 10.1  | 9.9   | 9.7   | 10.5  | 12.7  | 14.6  | 16.6  |
|                                                                        | 5% WB       | MCDBR  | 18.3  | 20.2  | 19.3  | 21.5  | 19.6  | 18.9  | 19.2  | 18.0  | 20.5  | 20.0  | 19.7  | 17.8  |
|                                                                        |             | MCWBR  | 17.8  | 17.2  | 14.7  | 14.7  | 10.5  | 9.9   | 9.9   | 9.4   | 9.9   | 12.2  | 15.0  | 17.2  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.303 | 0.326 | 0.351 | 0.387 | 0.422 | 0.443 | 0.453 | 0.437 | 0.384 | 0.358 | 0.322 | 0.303 |
|                                                                        | taud        |        | 2.430 | 2.349 | 2.355 | 2.278 | 2.210 | 2.190 | 2.179 | 2.210 | 2.336 | 2.402 | 2.469 | 2.461 |
|                                                                        | Ebn at Noon |        | 273   | 281   | 285   | 281   | 272   | 266   | 262   | 263   | 271   | 266   | 264   | 264   |
|                                                                        | Edh at Noon |        | 25    | 32    | 36    | 41    | 45    | 46    | 46    | 43    | 36    | 30    | 24    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 495   | 767   | 1084  | 1475  | 1733  | 1892  | 1862  | 1709  | 1386  | 941   | 626   | 421   |
|                                                                        | RadStd      |        | 48    | 100   | 104   | 135   | 146   | 138   | 110   | 84    | 123   | 82    | 88    | 50    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (49 neighbors) | +0.74 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +221  | +122  |

Nomenclature: See separate page